

Development of Electric Power Steering System for F-1 Race Car

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ABSTRACT

An electric power steering system for an F-1 car was developed and fitted into a 2000-2001 model F-1 car. The purpose of this development was to reduce steering effort, which can exceed 350 N during a race. The new system achieves the assist performance required for the F-1 car and is compact and lightweight, with low energy consumption. To achieve this, the motor, the motor control unit driving the motor, the torque sensor, the steering gear box containing the reduction gear and the control systems were all redesigned. During development, assist performance and response were evaluated on bench tests, and stroking durability and reliability were evaluated by simulation using actual car data. Final configuration was determined using track tests. As a result, a system was developed which meets assist performance targets and satisfies the F-1 car durability and reliability requirements.

1. Introduction

Current F-1 vehicles see a maximum steering effort of 350 N or more, equivalent to between 5 to 10 times that of mass produced vehicles. In a race that can last up to two hours, reducing the steering effort is essential to help the driver maintain both physical strength and concentration. Therefore, at present, all F-1 vehicles are fitted with power steering systems. Most other constructors use a hydraulic source, which is shared with the transmission control at approximately 18 MPa, and usually an electronically controlled flow control valve.

At Honda, during the initial spec study stage, a comparative study was made of hydraulic power steering system and electric power steering (EPS) system. The EPS system was selected for its low energy consumption. Figure 1 shows the energy consumption, calculated from steering torque according to actual running test data of hydraulic power steering and EPS. As shown in that graph, in

circuit driving, EPS gives a 16% better average energy consumption.

This paper provides an overview of the F-1 EPS system, and its performance.

2. Regulations

In the 2001 FIA (Fédération Internationale de l'Automobile) regulations concerning F-1 vehicle steering it states, "Power assisted steering systems are permitted but may not carry out any function other than reduce the physical effort required to steer the car." That is to say, it is prohibited to control vehicle action with the purpose of assisting the driver's driving competency through power steering control. It states, "The only permissible inputs are steering torque, steering angle, and car speed," and it is also prohibited to use differential values or integrated values of these input signals as control inputs. In addition, if these inputs are filtered, the

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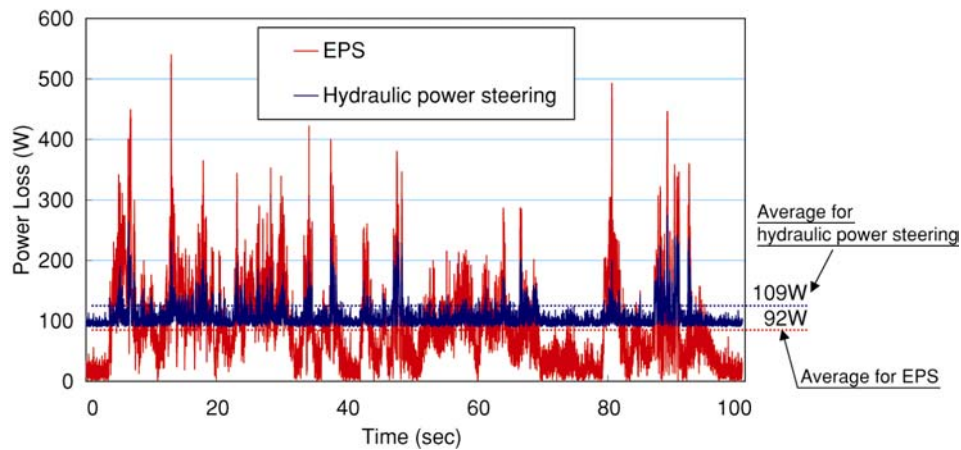


Fig. 1 Estimated power loss in hydraulic power steering and EPS; Suzuka circuit

cut-off frequencies must be approved and published by the FIA. That is to say, control that proceeds according to assist torque phase, using a band pass filter, is prohibited.

In 2002, the regulations were modified to state, "Power assisted steering systems may not be electronically controlled or electrically powered." Therefore, from 2002 it is no longer possible to use EPS on F-1 vehicles.

3. Development Concepts and Target Performance

In the case of mass production vehicles, power steering assist performance settings are generally made with the focus on static steering. However, with F-1 vehicles the down force changes according to the vehicle's designed aerodynamic characteristics. The steering holding effort has its greatest load during high speed cornering when the maximum tire vertical load is reached. Furthermore, there is a dynamic performance demand for power assistance in response to quick steering angle speeds when countersteering.

Currently, in the pursuit of aerodynamic performance, the F-1 vehicle body has become a sleek, spearhead-like mono-form shape. Therefore, it is essential that the EPS motor and reduction gear mechanism be as compact as possible to be compatible with the layout of the front suspension and brake system in the limited space available.

In addition, thorough weight reduction activity on the EPS unit is required to keep the weight increase due to installing the EPS unit as small as possible. Lastly, it is necessary to keep energy consumption to a minimum, yet have a design that makes a sufficient energy supply with the existing AC generator and battery.

To achieve this, the development concepts for the F-1 EPS were as follows.

- Compact and lightweight.
- Meet target assist torque and dynamic performance.
- No modification to the existing charge and discharge system.
- Meet durability and reliability levels so that there is no need to retire from the race due to an EPS failure.

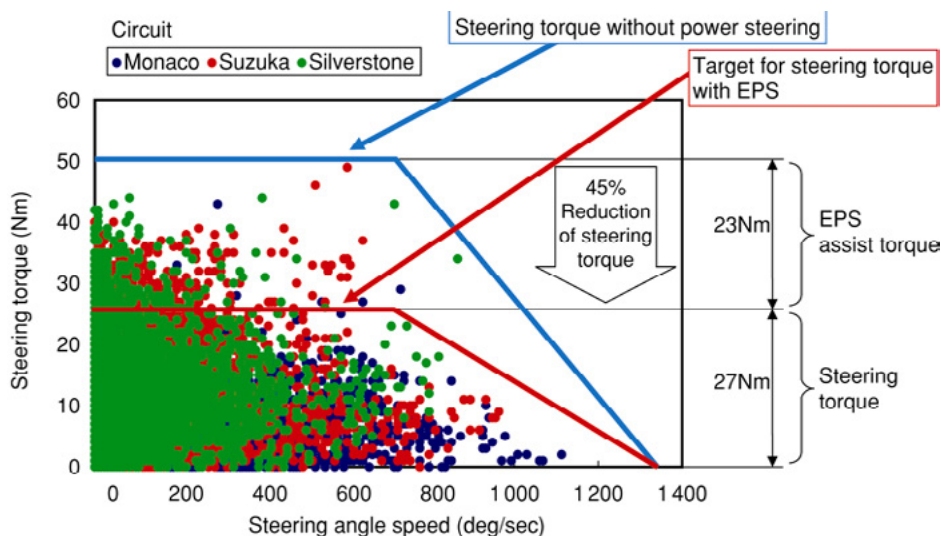


Fig. 2 Steering angle speed vs steering torque without power steering at each circuit

To set specific performance targets, a reference was made using results from trials of steering effort reduction using the EPS for mini-cars. This testing was carried out as part of past Japan Touring Car Championship (JTCC) race vehicle development. Based on these results, when steering effort is too light due to excessive assist, race vehicle controllability worsens. It was determined that steering effort of between 150 ~ 200 N is most appropriate. This level was used as the steering effort target in the current project. Through multiplication of the F-1 steering wheel diameter the target steering torque was then set at 20 ~ 27 Nm. Next, from track test data of past F-1 vehicles without power steering installed, a distribution of steering torque versus steering angle speed for each circuit was determined, as shown in Fig. 2. From this it was found that maximum steering torques of 50 Nm can occur at an angular steering rate of up to 720 deg/sec. Thus, in order to achieve the targeted steering

torques, there must be a 45% reduction in steering effort over the whole range. Therefore, the maximum EPS assist torque target was 23 Nm, and dynamic performance targets required that this assist torque be generated at a maximum angular steering rate of 720 deg/sec.

As for charging and discharging feasibility, an estimation was taken of the generated current by the AC generators on the circuit with the highest assist load. The current that can be consumed by the EPS is then estimated from the difference from this level to the current consumption of the electrical systems other than the EPS. Based on these estimates, the target established was 5 A or less as the average current consumption allowed to the EPS system per circuit.

4. System Overview

The components for the F-1 EPS are shown in Fig. 3. Figure 4 shows the EPS system block diagram.

The F-1 EPS system is comprised of the following.

- Torque sensor; detects the driver's steering torque and outputs the torque signal.
- ECU; receives the torque signal and outputs the assist torque instruction value.
- MCU; receives the assist torque instruction value from the ECU and controls motor torque.
- Steering gearbox; is integrated with a motor generating assist torque and a reduction gear mechanism which amplifies that drive torque.
- Vehicle speed sensor; detects vehicle speed.

4.1. Torque Sensor

In the case of a mass production vehicle's EPS, generally the torsion bar's torsion angle is mechanically converted to axial direction displacement. The differential transformer

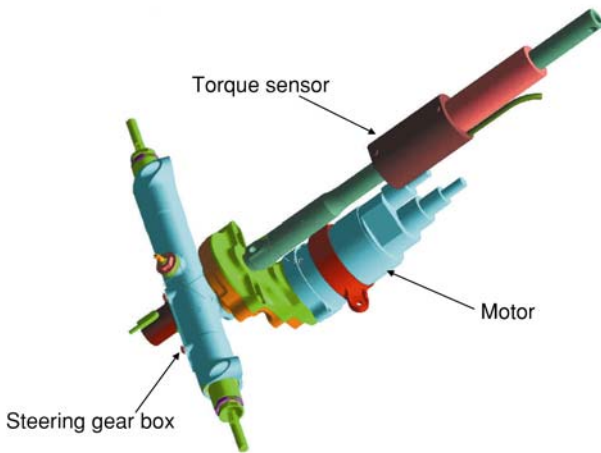


Fig. 3 EPS component for F-1 Car

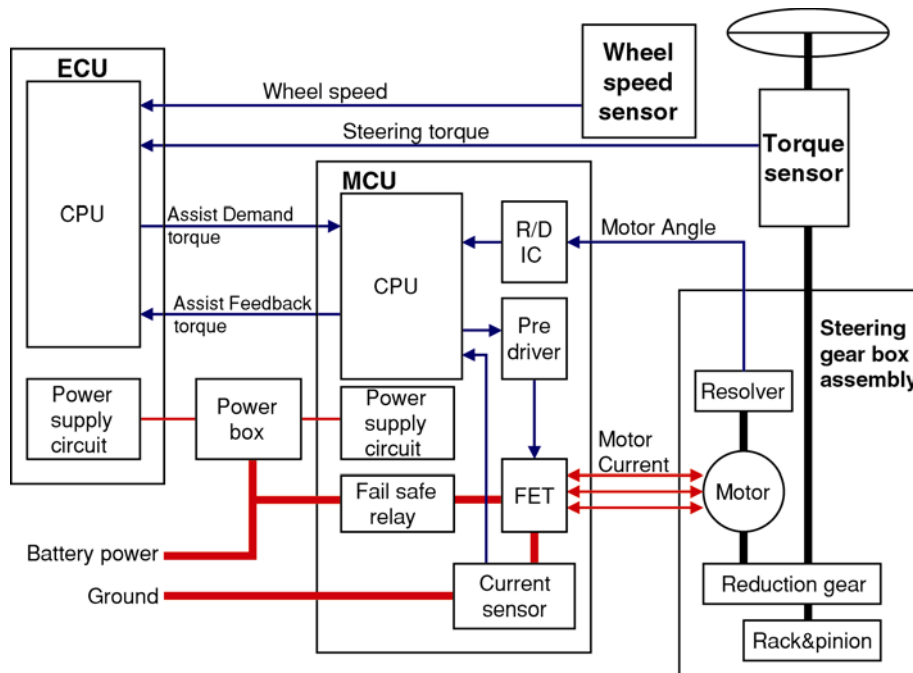


Fig. 4 EPS system block diagram

detects that displacement, then an electrical signal is output. With an F-1 EPS, as shown in Fig. 5, a strain gauge is directly mounted to the extension shaft that extends from the steering pinion, whereby the voltage of the bridge circuit, formed by the strain gauge, is detected – a simple structure with few parts. Focusing on ease of on-the-spot maintenance and ease of handling, there is only one component, a shaft with integrated strain gauge amp. In addition, in order to cope with the extreme vibrations generated by the F-1 vehicles which may reach as high as 30 G, the amp is mounted on floating rubber.

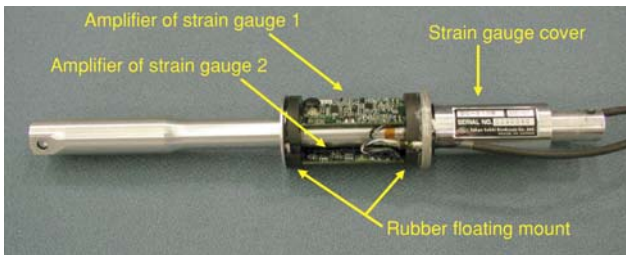


Fig. 5 Torque sensor

4.2. Motor

The motor selected was a specially developed DC brushless motor with advanced reliability and responsiveness. Figure 6 shows this motor. In order to make the motor smaller and more efficient, the magnets adopted were neodymium-based magnets. For ease of fitting to the F-1 vehicle, motor size was set at a diameter of 60 mm max with a length of 130 mm max. The aim was to achieve motor characteristics that satisfied both target assist performance and current consumption within these dimensions.



Fig. 6 EPS motor

4.3. Motor Control Unit (MCU)

At the same time, an MCU was developed to carry out torque control, through a drive circuit unique to the brushless motor. Figure 7 shows this MCU. In circuit driving mode, maximum assist torque frequently occurs, and a FET was selected that has sufficient capacity to withstand the generated heat. In addition, it has a motor control mechanism monitoring function and carries out mutual failure diagnosis through communication with the ECU.



Fig. 7 Motor control unit (MCU)

4.4. Steering Gearbox with Reduction Gear Mechanism

It was essential to make the reduction gear mechanism (that amplifies the motor torque) more compact, in order to maintain a layout inside the mono-form shape with the suspension and brake system parts. By using a titanium alloy helical gear in a double reduction gear mechanism, the external diameter shape can be reduced to 3/5 of that of a



Fig. 8 Steering gear box assembly

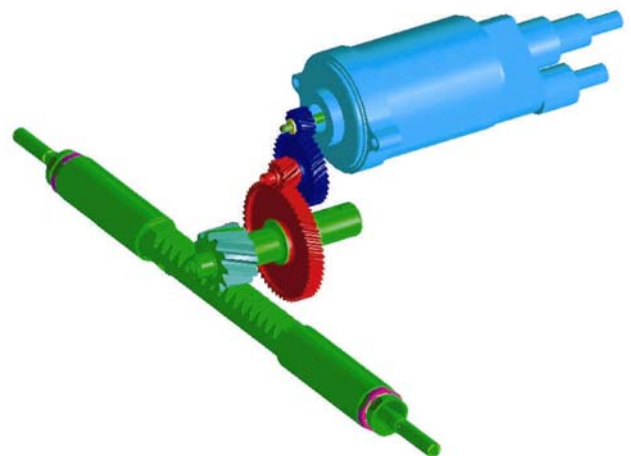


Fig. 9 Reduction gear

single reduction gear mechanism. Figure 8 shows the steering gearbox, and Fig. 9 shows the reduction gear mechanism. With this change, it is possible to fit the motor in parallel with the pinion shaft at the bottom of the gearbox, giving the vehicle a low center of gravity. In addition, two different types of reduction ratios have been designed, to match certain circuit characteristics. The actual rack and pinion steering mechanism itself is made of titanium alloy, and the gear housing is made from magnesium alloy, thereby reducing weight.

5. Control Specifications

With this F-1 EPS, the EPS assist torque as compared to the sum of the driver's steering torque and EPS assist torque, is set as the standard assist rate. However, in order to prevent system oscillation, a dead band was established in the steering torque neutral range. The standard assist rate R_a is defined as follows.

$$R_a = T_a / (T_s - T_d + T_a) \quad (1)$$

T_s : Driver steering torque
 T_a : EPS assist torque
 T_d : Dead band

In comparison with this standard assist rate, the assist rate is increased or decreased in response to vehicle speed according to coefficients specified in the speed control table. Final assist torque, T_v according to vehicle speed control is defined as follows.

$$T_v = R_a(T_s - T_d) \cdot K_v / (1 - R_a) \quad (2)$$

K_v : Vehicle Speed Coefficient

A standard assist rate and vehicle speed control table, as shown in Fig. 10, establishes five types of standard assist rates (0, 30, 35, 40, 45%) for each circuit, and gives the vehicle speed control table in response to each. These assist modes are stored in the ECU. The driver selects the mode, using an EPS mode switch on the steering wheel, to match the course and the driving conditions. However, because the

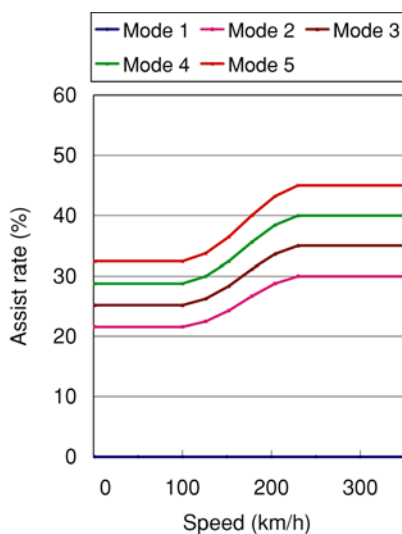


Fig. 10 Speed vs assist rate

regulations prohibit changing power steering characteristics while driving, mode switching can only be carried out while the vehicle is stationary.

6. System Reliability

The primary focus of the F-1 EPS system reliability is that an EPS malfunction must not be a cause of retiring from a race. Therefore, a feature was added so that the assist function stops within a specified time and manual steering function returns. Also, a switch was added so that the driver can interrupt the system at his/her discretion. Both are added in case a malfunction occurs or the appropriate assist function cannot be maintained.

In addition, there are instances where the assist torque demand exceeds the EPS estimated load, due to unpredictable changes in conditions. To maintain the assist function even in these situations, there is a protective function that monitors MCU drive circuit temperature and limits the amount of assist so that the tolerated temperature is not exceeded. There is also a protective function that limits the amount of assist in response to voltage drop, so that the power supply voltage can be maintained.

7. Durability and Reliability

With an F-1 vehicle, there are specified driving distances for which it is possible to use up a part, and replacement periods are managed correspondingly. For EPS, too, in order to establish the driving distance for which function can be guaranteed, stroke durability tests were carried out using a bench tester. This bench reproduces rack input loads and steering angles from actual running test data. Vibration resistance tests were also carried out, reproducing actual car vibration levels. The driving distance specified in these tests were verified with track tests in the end, and as long as there are no problems, the parts will be managed based on these specifications.

In addition, all EPS components are fitted inside the driver's cockpit. The EPS motor, sensors, and electrical units have all been given a waterproofing treatment, to protect them from water permeation into the cockpits when driving in rain.

8. Results

8.1. Weight

Shown in Table 1 are the results of a thorough weight reduction activity giving a total EPS weight of 4.09 kg, and a total weight increase due to adding EPS of 3.35 kg.

Table 1 Weight of EPS (kg)

	Original steering	EPS
Torque sensor	0.1	0.35
Motor		1.4
MCU		0.9
Harness		0.5
Steering gear box	0.64	0.94
Total weight	0.74	4.09

8.2. Performance

Table 2 gives the general specifications for the F-1 EPS.

Performance of the EPS fitted to the 2002 model realizes dynamic characteristics of maximum assist torque of 23.4 Nm up to a maximum steering angle speed of 720 deg/sec. It was possible to achieve both the F-1 required assist torque and the dynamic performance.

Table 2 Specifications of EPS

Type of EPS	Pinion assist type
Rated electric voltage	12V
Rated electric current	15A
Maximum assist torque	23.4Nm
Dynamics	4rev/sec
EPS reduction gear type	Double Helical gear
EPS reduction gear ratio	10.6
	13
Type of Motor	DC brushless Motor
Maximum Motor Torque	1.8Nm
Inertia of Motor	0.004g·m ²
Electric time constant	0.7msec
Type of Torque sensor	Strain gauge type

8.3. Current Consumption

Figure 11 shows the 2001 F-1 Japanese Grand Prix Suzuka circuit's actual running data.

In order to reduce the average EPS current consumption, an assist torque maximum value was established. The MCU carried out current consumption upper limit value control in response to motor speed, so that the maximum torque is not exceeded.

Actual running test results showed that on the Suzuka circuit with the largest assist torque load, the average consumed current for one circuit was kept to 3 A, which is below the target.

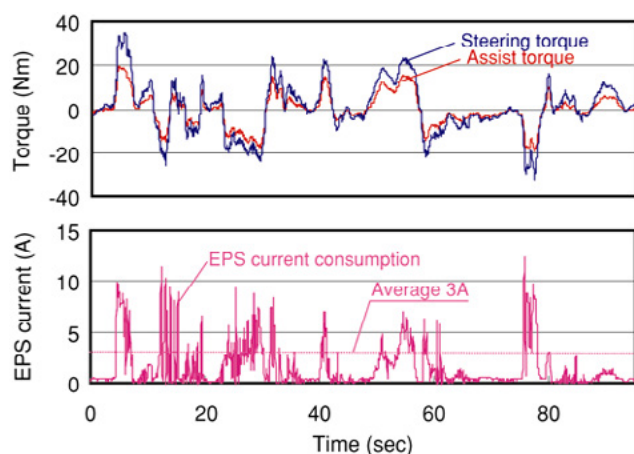


Fig. 11 Results for EPS current consumption (Suzuka)

9. Summary

A small, lightweight EPS system for F-1 vehicles was developed, which achieves both the required assist performance and energy consumption requirements. This

EPS was fitted to the 2000 ~ 2001 model F-1 vehicle. With this EPS, it is possible for driver steering effort, in response to each circuit's characteristics, to be reduced by up to 45%.

10. Conclusion

By fitting an F-1 EPS system, it is possible to improve the ease of steering when cornering, to improve drivability, and, as noted by the drivers, to reduce physical exhaustion in a race, helping to maintain driver concentration. It is clear that this is an effective F-1 race vehicle system.

11. Acknowledgements

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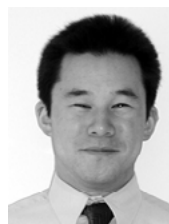
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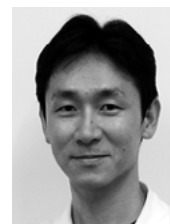
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